

ST-MGPrompt

Overall framework · train-only priors

Historical input



Historical wind-farm data

134 turbines
16 variables
144 steps / 24 h

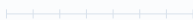
X in $\mathbb{R}^{(B \times 144 \times 134 \times 16)}$

16 → 64

shared projection

1 Dual Temporal Semantics

one 144-step timeline · same grid



Fine



Coarse



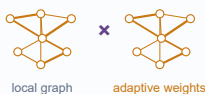
Fine / Coarse = 1/2 + 1/2

2 Graph-Temporal Encoding

same support · different semantic evidence

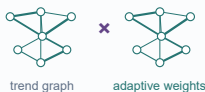
Micro

$\Delta W_{spd} + \text{distance}$



Macro

Patv trend + distance

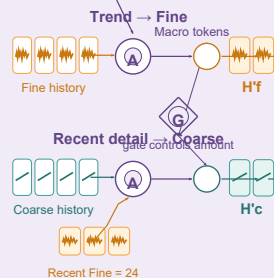


3 Cross-Granularity Fusion

Node-specific macro-trend prompt

P1 P2 P3 P4

$P = 4 \text{ tokens / turbine}$



$$H^f = \text{LN}(H_f + G \cdot O_{m \rightarrow f})$$
$$H^c = \text{LN}(H_c + (1-G) \cdot O_{f \rightarrow c})$$

4 Forecasting Context

Context

last-state + macro mean

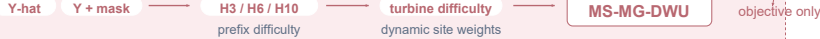
context vector

[node] [horizon] [type]
Node-Horizon ST Prompt

Direct decoder · H=10

future horizon
 $\hat{Y}$ in $\mathbb{R}^{(B \times 10 \times 134)}$
H3 H6 H10

Training only · MS-MG-DWU



optimization branch; no forward-path input